ATMA: Long-Context Language Modeling via Polar Attention and Gated-Delta Compression Memory

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Abstract

Zero-shot length extrapolation requires a sequence model to balance exact retrieval, long-document likelihood, and inference cost. We introduce ATMA, a hybrid architecture whose Polar Attention separates a radially normalized direction channel from a bounded participation-ratio magnitude channel and augments both with gated-delta recurrent memory. We evaluate long-context modeling as a Pareto problem rather than claiming universal architectural dominance. A 120-run, 1B-token factorial sweep selects the recipe; matched 10B-token, 378M-parameter NoPE, RoPE, and Polar models, plus stateful Raven baselines, are then evaluated through 256K tokens. Polar preserves real-text retrieval at 64K (16.3% versus 0.0% for NoPE and RoPE) and limits 256K bits-per-byte inflation to 1.26×, with a 1.5-point mean penalty on short-context tasks. Raven instead leads 256K BABILong and offers constant-state decoding. Finally, nominally matched NoPE runs have indistinguishable 2K validation curves yet diverge sharply at 256K across GPU device classes. The results motivate multi-objective and multi-environment audits of long-context claims.

1 Introduction

Language models increasingly process repositories, books, and scientific documents far longer than their training sequences. Standard softmax attention retains explicit access to old tokens, but its dense normalization can disperse mass as the key population grows (Nakanishi, 2025; Velickovic et al., 2025; Barbero et al., 2024). A complementary extreme-value view observes that the largest chance match among unrelated keys grows with context length (Huang et al., 2026). Sliding windows bound the active population but discard distant targets; recurrent mixers keep fixed state but compress the past. Long-context modeling therefore presents several competing objectives: retrieval fidelity, document-level likelihood, reasoning, short-context quality, and serving cost.

We introduce ATMA, a 378M-parameter hybrid decoder that combines local gated convolutions with four global Polar Attention layers. Polar adds a learned null competitor that grows on the $\sqrt{\log n}$ extreme-value scale, radially normalizes the matched value sum, and exposes effective match count through a bounded participation-ratio channel. A gated-delta memory supplies a third, recurrent channel. The design separates "what matched," "how much matched," and compressed history while keeping every channel bounded under explicit conditions (Appendix A).

Our study makes three contributions. First, a complete 120-cell, 1B-token factorial sweep isolates the interaction between attention core, recurrent memory, local windows, distractor alignment, and regularization. Second, matched 10B-token attention runs and strong Raven baselines are compared on retrieval, BABILong, fixed-target likelihood, eight short-context tasks, and serving. Figure 1 summarizes the resulting Pareto frontier: Polar leads the attention variants in long-context retrieval and stability, Raven leads 256K BABILong and decode efficiency, and softmax retains a small short-context advantage. Third, a controlled diagnostic finds that NoPE checkpoints with effectively identical 2K training curves can differ by 6.70 nats at 256K after training on different GPU classes. We report this as

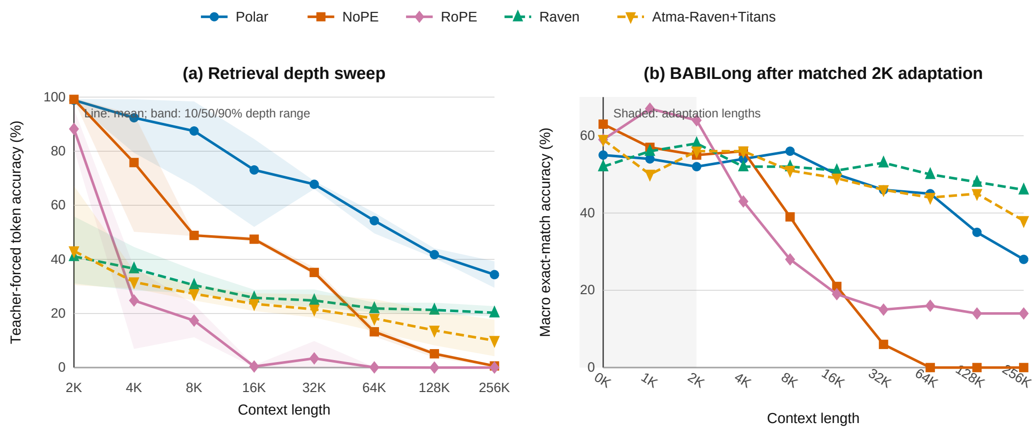

Figure 1: Matched 10B-token results. Left: teacher-forced retrieval accuracy, averaged over retrieval tasks, haystack suites, and needle depths; bands span depth-wise means. Right: BABILong macro exact match after identical answer-only adaptation. Polar leads matched attention variants at long lengths; Raven leads long-context reasoning but changes both architecture family and optimizer.

evidence of environment sensitivity, not as proof that reduction order is the unique cause, and call for multi-device replication before architectural attribution.

2 Related Work

Softmax attention (Vaswani et al., 2017) has quadratic cost and may lose sharpness as the number of competitors increases. RoPE (Su et al., 2021) and ALiBi (Press et al., 2021) alter positional treatment but not dense simplex normalization. Scalable-Softmax, adaptive-temperature analyses, and sparse alternatives explicitly study dispersion or size-dependent sharpness (Nakanishi, 2025; Velickovic et al., 2025; Vasylenko et al., 2026). Threshold Differential Attention (TDA) instead uses a $\sqrt{\log n}$ threshold motivated by maxima of noise scores (Huang et al., 2026). Polar adopts this scale but uses a differentiable null sink rather than hard survivor selection.

Linear attention, state-space models, and gated recurrences exchange explicit token access for compressed state (Gu & Dao, 2023; Yang et al., 2024b;a; Behrouz et al., 2025). Raven combines routed sparse memory, decay dynamics, and gated-delta recurrence (Afzal et al., 2026); we treat it as a strong sub-quadratic baseline, but not as an attention-only ablation because its optimizer and model family differ. Hybrid models combine local and global mixers for efficiency (Hasani et al., 2025; Allen-Zhu, 2025). ATMA follows this hybrid pattern while changing the global readout and adding a recurrent memory channel.

3 Method

3.1 Hybrid architecture

ATMA is a 16-layer decoder with 12 LFM2-style gated convolutional layers and four global attention layers. All blocks use pre-normalization and squared-ReLU MLPs. The matched NoPE, RoPE, and Polar variants share hidden size 1024, eight query heads, two key-value heads, head dimension 128, tokenizer, data order, optimizer, schedule, and training budget; only the attention core changes. Canon depthwise causal convolutions are applied to q, k, and v before scoring (Allen-Zhu, 2025).

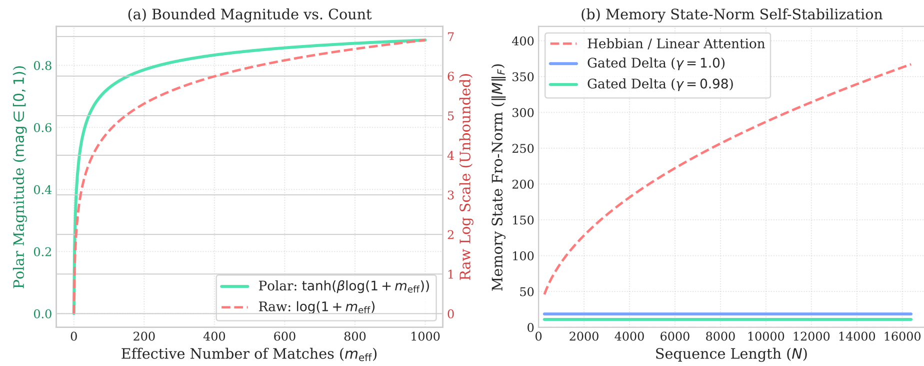

Figure 2: ATMA mixer diagnostics. Left: the effective match count is mapped to a bounded magnitude channel. Right: the tested gated-delta state remains flat with length, unlike an ungated Hebbian memory. These controlled diagnostics illustrate the mechanism; they are not a network-level stability proof.

3.2 Polar attention

For query i and its $n_i = i + 1$ visible keys, let $\sigma_{ij} = q_i^\top k_j / \sqrt{d_k}$. Each head learns a length temperature and a virtual null logit,

$$t_i = 1 + \mathrm{softplus}(\alpha) \log n_i, \qquad \nu_i = b + \mathrm{softplus}(\gamma)\sqrt{\log(n_i + 1)}. \tag{1}$$

The second term tracks the extreme-value scale of unrelated scores. Appending the null as a softmax competitor gives

$$w_i = \mathrm{softmax}(t_i[\sigma_{i\bullet}, \nu_i]), \qquad s_i = \sum_j w_{ij} v_j + w_{iN} v_{\text{null}}. \tag{2}$$

The direction channel is $c_i = s_i / \max(\|s_i\|_2, \varepsilon)$, hence $\|c_i\|_2 \le 1$. To retain bounded information about match multiplicity, real-key weights are renormalized as $\hat{w}_{ij} = w_{ij} / \sum_k w_{ik}$ and summarized by the participation ratio

$$n_{\text{eff},i} = \left(\sum_j \hat{w}_{ij}^2\right)^{-1}, \quad m_{\text{eff},i} = n_{\text{eff},i}(1 - w_{iN}), \quad \mu_i = \tanh(\mathrm{softplus}(\beta)\log(1 + m_{\text{eff},i})). \tag{3}$$

Thus $\mu_i \in [0, 1)$, while the direction channel removes radial scale. This is a channel decomposition, not an independence claim: changing the match set can rotate c_i. The Polar output is a gated projection of c_i plus a learned projection of μ_i. A distractor-alignment auxiliary loss was tested but is disabled in the selected recipe because the factorial sweep shows worse retrieval.

3.3 Gated-delta memory

Each global layer also maintains a per-head associative state $M_t \in \mathbb{R}^{d_v \times d_k}$. With unit-normalized keys and queries, retention γ_t, and write gate β_t, the decay-first, self-inclusive recurrence is

$$M_t = \gamma_t M_{t-1}(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top, qquad r_t = M_t q_t. \tag{4}$$

RMS-normalized r_t is gated, projected, and added to the Polar output. The projection is zero-initialized. Unit L_2 normalization is important: RMS-normalized keys have $\|k\|_2^2 = d_k$ and made the tested delta recurrence diverge, whereas unit keys keep the update eigenvalue along k within the gate-controlled range. Under bounded values and a uniformly contractive retention gate, Appendix A gives a sequence-length-independent state bound. Our empirical claim is narrower: the implemented state remains stable in the tested length sweep.

1B recipe	NLL 2K	NLL 64K	Needle 64K
NoPE + memory	2.665	2.342	16.3%
RoPE + memory	2.808	2.857	0.0%
Polar, no memory	2.826	3.604	0.0%
Polar + memory	2.703	1.960	92.5%
Polar + memory + distractor	2.725	1.975	58.8%
Polar + memory + window	2.712	2.006	30.0%

Table 1: Illustrative 1B-token recipe selection. NLL is nats/token; Needle is greedy per-digit accuracy over 16 five-digit trials.

4 Experimental Design

Recipe selection. Stage I trains all $3 \times 5 \times 2 \times 2 \times 2 = 120$ combinations of attention type (Polar, RoPE, NoPE), regularization, distractor loss, memory, and a 1024-token window. Models receive approximately 1B FineWeb-Edu tokens at length 2048 on NVIDIA L4 GPUs. This stage selects a recipe; it is not used for headline absolute comparisons.

Matched comparison. Stage II trains 378.16–378.22M-parameter NoPE, RoPE, and Polar models for 9.816B tokens at length 2048 on the same L40S software and hardware environment. All use the same Muon/AdamW split. Raven Native (388.54M) and Atma-Raven-Titans (382.37M) use the same data, tokenizer, budget, and device class, but Raven's native AdamW optimizer; they form a separate baseline group.

Evaluation. The suite contains 24,000 paired retrieval trials over synthetic and FinePDFs haystacks, lengths 2K–256K, depths 10/50/90%, and 50 trials per cell; BABILong answer-only adaptation and held-out exact match (Kuratov et al., 2024); fixed-target bits per byte (BPB) on FinePDFs, PG-19, and Proof-Pile; eight likelihood-scored short-context tasks (22,939 examples per model); and single-sequence prefill/decode on one L40S. Retrieval is teacher-forced five-token accuracy, which is less brittle than exact-value match but should not be read as free-running generation success. BABILong uses a custom split and prompts, so its absolute scores are internally comparable rather than leaderboard replications. Protocol details and complete matrices appear in Appendix B.

5 Results

5.1 Component selection

Table 1 shows the decisive transitions. Neither memoryless Polar nor memory-augmented softmax retrieves reliably at 64K. Combining Polar with gated-delta memory raises per-digit retrieval from 0.0% to 92.5%; a local window or distractor alignment then reduces it. We therefore promote global Polar plus memory without the auxiliary distractor loss. The complete 120-cell matrix is in Appendix C.

5.2 A multi-objective frontier

Table 2 reports matched endpoints. At 2K, NoPE and Polar both retrieve almost perfectly. By 256K, NoPE and RoPE fall to 0.6% and 0.0%, while Polar retains 34.4% averaged across tasks, suites, and depths. The aggregate hides an important boundary: on natural FinePDFs haystacks, Polar retains 16.3% at 64K versus 0.0% for NoPE and RoPE, but falls to 0.3% at 256K; on synthetic filler it retains 68.4%. We therefore claim a substantially extended effective window, not solved arbitrary-length retrieval. Full depth and haystack matrices are in Appendix D.

Polar also limits mean BPB inflation to 1.26× at 256K, compared with 5.65× for NoPE and 1.75× for RoPE. This stability accompanies a short-context cost: mean accuracy over eight 2K controls is 43.29% for Polar, 45.15% for NoPE, and 44.60% for RoPE. The 1.86-point

Group	Model	Retrieval (%)		BABILong (%)		BPB		
		2K	256K	2K	256K	2K	256K	ratio
Matched	NoPE	99.2	0.6	55	0	1.432	8.097	5.65×
Matched	Polar	98.9	34.4	52	28	1.451	1.825	1.26×
Matched	RoPE	88.2	0.0	64	14	1.439	2.512	1.75×
Raven/AdamW	Atma-Raven-Titans	43.1	10.0	56	38	1.525	1.551	1.02×
Raven/AdamW	Raven Native	41.1	20.3	58	46	1.581	1.597	1.01×

Table 2: Long-context endpoints. Retrieval averages tasks, suites, and depths. BABILong is macro exact match. BPB averages three fixed-target datasets; lower is better. Bold is best within each comparison group.

Model	Haystack	2K	4K	8K	16K	32K	64K	128K	256K
Polar	Synthetic	99.6	98.4	98.5	97.0	94.9	92.3	80.3	68.4
	FinePDFs	98.6	86.3	77.9	49.1	40.7	16.3	3.2	0.3
NoPE	Synthetic	99.5	98.7	98.9	94.9	70.3	26.5	10.2	1.2
	FinePDFs	99.3	52.9	0.0	0.0	0.0	0.0	0.0	0.0
RoPE	Synthetic	81.7	44.9	33.1	0.8	6.8	0.1	0.0	0.0
	FinePDFs	94.8	4.7	0.2	0.0	0.0	0.0	0.0	0.0
Raven Native	Synthetic	55.9	57.7	51.4	48.3	44.3	40.9	38.9	35.1
	FinePDFs	22.1	15.4	8.2	3.3	5.3	2.9	3.7	5.4

Table 3: Zero-shot retrieval accuracy (%) by haystack type. Values average teacher-forced token accuracy over needle depths. The recurrent hybrid is omitted for space and reported in Appendix D.

gap to NoPE (approximately 1.5 points relative to the matched-group mean) prevents a blanket quality claim. Per-task results are in Appendix E.

Raven exposes a different trade-off. It reaches 46% BABILong at 256K and keeps BPB almost flat, while its fixed recurrent state avoids length-dependent KV-cache reads. At 128K, Atma-Raven-Titans decodes at 2.27 ms/token versus 16.3 ms/token for Polar, but Polar has higher synthetic retrieval and higher training MFU. Because Raven also changes optimizer, these results locate an operational frontier rather than isolate one mechanism.

5.3 Environment sensitivity

Figure 3 shows the most consequential diagnostic. Three NoPE runs finish at 2.8126, 2.8128, and 2.8160 validation nats, yet their 256K NLLs are 10.77 (L40S, microbatch 16), 9.90 (L40S, microbatch 4), and 4.07 (L4, microbatch 4). Re-evaluation under two PyTorch versions did not change scores, and matching microbatch size on L40S did not recover the L4 behavior. All five headline Stage II models were trained together on L40S, so the main comparison remains environment-matched.

Group	Model	Base mean (%)	128K prefill (s)	Decode (ms/token)	Peak alloc. (GiB)	Train MFU (%, 2K)
Matched	NoPE	45.1	1.47	16.4	39.40	42.77
Matched	Polar	43.3	1.62	16.3	39.41	40.44
Matched	RoPE	44.6	1.47	16.5	39.41	42.45
Raven/AdamW	Atma-Raven-Titans	43.1	1.00	2.27	8.62	30.92
Raven/AdamW	Raven Native	43.0	1.36	3.27	6.46	21.53

Table 4: Short-context and systems trade-offs. Serving uses one sequence and one timing sample on one L40S, so the values are descriptive rather than uncertainty estimates.

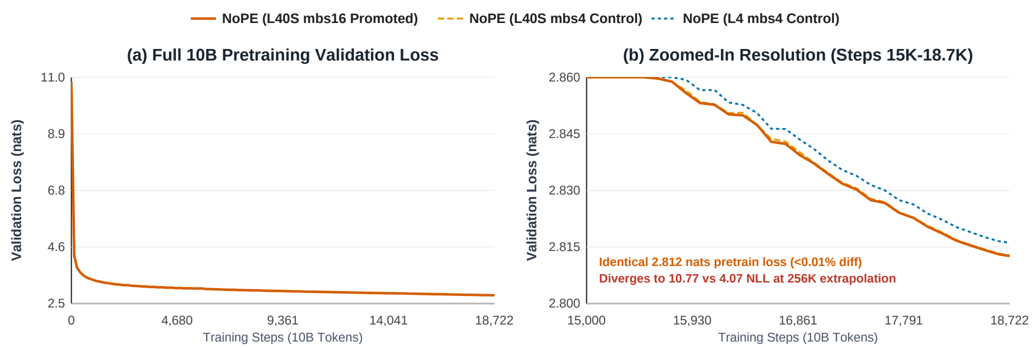

Figure 3: NoPE 2K validation trajectories are nearly coincident, yet the checkpoints' 256K NLL values are 10.77, 9.90, and 4.07. The two L40S runs differ in microbatch size; the L4 run differs in device class.

FEA Structural Stress Probing: Local Block Secant Gains G Across Length (2K -> 256K)

Polar (ATMA): Flat Gains Across Length

	2K	4K	8K	16K	32K	64K	128K	256K
B0	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.86
B1	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78
B2	0.96	0.95	0.95	0.95	0.96	0.96	0.96	0.96
B3	0.90	0.89	0.89	0.89	0.88	0.88	0.88	0.87
B4	0.87	0.87	0.86	0.86	0.86	0.85	0.85	0.85
B5	0.86	0.85	0.85	0.84	0.84	0.84	0.84	0.84
B6	1.02	1.02	1.03	1.05	1.07	1.09	1.10	1.11
B7	0.98	0.97	0.97	0.97	0.97	0.97	0.97	0.97
B8	0.96	0.96	0.95	0.95	0.95	0.95	0.95	0.94
B9	0.93	0.92	0.92	0.92	0.92	0.92	0.92	0.91
B10	1.10	1.10	1.10	1.10	1.10	1.10	1.10	1.09
B11	1.02	1.02	1.02	1.02	1.02	1.01	1.01	1.01
B12	1.02	1.01	1.01	1.01	1.01	1.00	1.00	1.00
B13	1.01	1.01	1.00	1.00	1.00	1.00	1.00	1.00
B14	1.43	1.44	1.45	1.44	1.41	1.39	1.38	1.37
B15	1.23	1.23	1.22	1.21	1.20	1.19	1.18	1.17

RoPE: Representational Attenuation

	2K	4K	8K	16K	32K	64K	128K	256K
B0	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94
B1	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92
B2	1.01	1.00	0.99	0.98	0.98	0.98	0.98	0.98
B3	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02
B4	1.05	1.07	1.08	1.07	1.06	1.04	1.03	1.03
B5	0.94	0.93	0.93	0.93	0.93	0.93	0.93	0.92
B6	1.09	1.06	1.04	1.02	1.01	1.01	1.00	1.00
B7	1.01	1.01	1.00	1.00	1.00	1.00	0.99	0.99
B8	1.01	1.00	1.00	1.00	0.99	0.99	0.99	0.99
B9	0.98	0.98	0.98	0.97	0.97	0.97	0.97	0.96
B10	1.09	1.08	1.07	1.06	1.06	1.05	1.04	1.04
B11	1.02	1.02	1.02	1.01	1.01	1.01	1.00	1.00
B12	1.02	1.02	1.01	1.01	1.01	1.00	1.00	1.00
B13	1.02	1.01	1.01	1.00	1.00	1.00	1.00	1.00
B14	1.32	1.31	1.29	1.28	1.27	1.26	1.25	1.25
B15	1.21	1.20	1.19	1.19	1.18	1.17	1.17	1.16

NoPE (L40S mbs16): Block 7 Yield Locus

	2K	4K	8K	16K	32K	64K	128K	256K
B0	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91
B1	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88
B2	1.02	1.02	1.02	1.02	1.02	1.03	1.03	1.03
B3	0.98	0.98	0.98	0.97	0.97	0.97	0.97	0.97
B4	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.95
B5	0.94	0.94	0.94	0.94	0.95	0.96	0.96	0.96
B6	1.13	1.11	1.09	1.09	1.08	1.09	1.09	1.10
B7	1.02	1.02	1.02	1.03	1.09	1.20	1.31	1.38
B8	1.01	1.01	1.01	1.00	1.00	1.00	1.00	0.99
B9	0.99	0.99	0.99	0.98	0.98	0.98	0.98	0.99
B10	1.16	1.14	1.12	1.12	1.12	1.11	1.11	1.11
B11	1.04	1.03	1.03	1.02	1.01	1.01	1.00	1.00
B12	1.04	1.04	1.03	1.02	1.01	1.01	1.01	1.00
B13	1.04	1.04	1.06	1.08	1.09	1.07	1.06	1.06
B14	1.39	1.37	1.35	1.35	1.30	1.26	1.24	1.23
B15	1.18	1.18	1.17	1.14	1.12	1.11	1.09	1.08

Figure 4: Sampled block secant gains across context length. The probe localizes the L40S NoPE checkpoint's change near blocks 6–7, while Polar remains comparatively flat. A sampled gain is a directional diagnostic, not a tight global Lipschitz estimate.

The evidence establishes sensitivity to the training environment, but it does not uniquely identify CUDA reduction order as the causal mechanism: device class is not randomized or replicated across multiple seeds, and other hardware-dependent kernels may contribute. The defensible conclusion is that short-context convergence did not predict long-context behavior in these runs. Architecture-level extrapolation claims should therefore report seeds, kernels, device classes, and extended-context audits; stronger causal claims require replicated cross-device interventions. Block-level stress maps and diagnostic controls are in Appendix F.

6 Limitations and Conclusion

The study uses one model scale, one 2K pretraining regime, and fewer independent training replicates than needed for population-level uncertainty. Teacher-forced retrieval, custom BABILong adaptation, and single-sample serving measurements answer different questions and should not be collapsed into one score. Polar still fails on natural-text retrieval at 256K, its theoretical bounds require learned-margin assumptions that are not yet empirically certified for every head, and Raven is not an optimizer-matched ablation.

Within those limits, ATMA demonstrates that bounded full-context attention plus recurrent memory can move the long-context Pareto frontier. Polar materially extends soft retrieval and stabilizes long-document likelihood while retaining competitive short-context quality; Raven illustrates the reasoning and serving advantages of compressed state. The environment diagnostic further shows why long-context claims need evaluation well beyond the training window and, where feasible, across hardware environments. We release configurations, logs, checkpoint manifests, benchmark matrices, and proof artifacts to make each reported comparison auditable.

AI Use Statement

Generative AI tools were used in this revision to edit and shorten the manuscript, condense the presentation of existing mathematical claims and proofs, improve readability, and suggest document structure. They were not used to generate experimental datasets or execute experiments in this revision. The authors will manually compare all AI-assisted mathematical text with the original derivations and proof artifacts, verify retained empirical claims against source logs and cited work, and take responsibility for every claim and artifact in the submission. This statement describes the present revision only; before submission, the authors will disclose any additional AI-assisted research tasks from the complete project history as required by the ICLR 2027 policy.

Reproducibility Statement

Model definitions and training settings are described in Sections 3–4. Appendix B specifies evaluation protocols and denominators; Appendices C–F provide detailed tables and diagnostics. The accompanying anonymous artifact contains configurations, logs, checkpoint manifests, benchmark payloads, and machine-checked proof skeletons.

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A Theoretical Framing

These are sufficient-condition results. They establish noise-odds and norm bounds under explicit assumptions; they do not establish distributional invariance or replace the empirical evidence.

A.1 Extreme-value null floor

Assumption 1 (Sub-Gaussian noise scores). For irrelevant keys $j \in \mathcal{N}_n$, $|\mathcal{N}_n| \le n$, scores X_j are mean-zero and v^2-sub-Gaussian: $\mathbb{E}\exp(tX_j) \le \exp(v^2t^2/2)$ for all $t \in \mathbb{R}$.

Let the Polar null floor be $\nu_n = b + s\sqrt{\log(n+1)}$ with $b, s \ge 0$.

Proposition 1 (Noise exceedances). Under Assumption 1,

$$\mathbb{E}\left[\sum_{j\in\mathcal{N}_n} \mathbf{1}\{X_j > \nu_n\}\right] \le n\exp\left(-\frac{\nu_n^2}{2v^2}\right). \tag{5}$$

If $s \ge \sqrt{2}v$ this is $O(1)$; if the inequality is strict it decays polynomially.

Proof. Apply the sub-Gaussian tail bound to each key and sum. Since $b \ge 0$, $\nu_n^2 \ge s^2\log(n+1)$. ☐

Unlike TDA, Polar retains sub-threshold keys. With temperature $\theta_n = 1 + a\log n$, define the unnormalized noise-to-null odds $R_n = \sum_{j\in\mathcal{N}_n}\exp(\theta_n(X_j - \nu_n))$.

Proposition 2 (Vanishing null odds). Suppose $s > c > \sqrt{2}v$ and the null floor is $s\sqrt{\log(n+1)}$ up to a nonnegative offset. With probability tending to one,

$$R_n \le n\exp\Big(-(1+a\log n)(s-c)\sqrt{\log n}\Big). \tag{6}$$

For $a > 0$, the bound converges to zero faster than any inverse polynomial.

Proof. A union bound gives $\Pr[\max_j X_j > c\sqrt{\log(n+1)}] \le n(n+1)^{-c^2/(2v^2)} \to 0$. On the complementary event, every noise score is at least $(s-c)\sqrt{\log n}$ below the floor. Bound each summand and sum over at most n keys. ☐

A.2 Bounded channels and recurrent state

Proposition 3 (Bounded Polar readout). For $c_i = s_i/\max(\|s_i\|_2, \varepsilon)$, $\|c_i\|_2 \le 1$. For $\mu_i = \tanh(\beta\log(1+m_{\mathrm{eff},i}))$, $\mu_i \in [0,1)$ when $\beta, m_{\mathrm{eff},i} \ge 0$. A distribution uniform over m real keys has participation ratio m.

Proof. The first two claims follow from the denominator and range of tanh. Uniform weights satisfy $(\sum_{j=1}^{m}(1/m)^2)^{-1} = m$. $\square$

Assumption 2 (Contractive memory). For every t, $\|q_t\|_2 = \|k_t\|_2 = 1$, $\|v_t\|_2 \le V$, $0 \le \beta_t \le 1$, and $0 \le \gamma_t \le \bar{\gamma} < 1$.

Proposition 4 (Length-bounded state). For recurrence equation 4,

$$\|M_t\|_F \le \bar{\gamma}^t \|M_0\|_F + \frac{V}{1-\bar{\gamma}}. \tag{7}$$

Proof. $I - \beta_t k_t k_t^\top$ has spectral norm at most one and $\|\beta_t v_t k_t^\top\|_F \le V$. Thus $\|M_t\|_F \le \bar{\gamma}\|M_{t-1}\|_F + V$; unrolling gives the result. $\square$

The Lean artifact checks the finite-sum, exponential-monotonicity, bounded-channel composition, and invariant-set induction steps. It takes the sub-Gaussian tail, high-probability margin, and one-step contraction as premises. It is therefore a machine-checked proof skeleton, not end-to-end formal verification. The learned conditions $b \ge 0$ and $s > c > \sqrt{2}v$ require an empirical head-wise audit before they can be asserted for a trained checkpoint.

B Evaluation Protocols

Stage I contains 120 approximately 370–378M-parameter runs trained for 1,900 steps of 524,288 tokens at sequence length 2,048 on FineWeb-Edu. Stage II contains three matched attention models trained for 18,722 such steps (9.816B tokens). The five promoted checkpoints and ten open-weight controls are evaluated with checkpoint-exact forward paths.

Retrieval crosses passkey and NIAH tasks, synthetic and FinePDFs haystacks, eight lengths from 2K through 256K, three depths, and 50 paired trials per cell. The primary metric is teacher-forced accuracy over five target tokens. BABILong uses answer-only fine-tuning on tasks qa1–qa10 at 0K/1K/2K and greedy exact match on ten held-out rows per task and length. Base-task controls use LAMBADA, HellaSwag, PIQA, WinoGrande, ARC-Easy, ARC-Challenge, OpenBookQA, and BoolQ. Fixed-target BPB uses FinePDFs, PG-19, and Proof-Pile. All reported processes completed without failed, OOM, unsupported, missing, or null cells.

C Complete Factorial Sweep

The complete 120-cell table is retained in the submission source and artifact. The main paper reports only the recipe transitions needed to justify promotion.

Table 5: Complete Results of the 120-Run Ablation Sweep. All models are 370M parameters, trained on FineWeb-Edu for 1B tokens (*seq_len* = 2048). Evaluated from 2K to 64K context length. Clean NLL is measured on coherent FinePDFs documents; junk NLL is measured on the concatenated validation stream. NLL is in nats/token (lower is better), Needle is greedy per-digit accuracy in % (higher is better), and MFU is training model flops utilization in %.

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
POLAR	baseline	Off	Off	Off	3.323	2.83	3.60	3.27	5.68	96.3	0.0	40.1
POLAR	baseline	Off	Off	On	3.343	2.83	2.66	3.30	4.74	67.5	0.0	39.6
POLAR	baseline	Off	On	Off	3.169	2.70	1.96	3.11	3.13	91.3	92.5	35.6
POLAR	baseline	Off	On	On	3.174	2.71	2.01	3.12	3.13	51.3	30.0	35.6
POLAR	baseline	On	Off	Off	3.332	2.81	5.56	3.28	6.54	85.0	0.0	33.3
POLAR	baseline	On	Off	On	3.361	2.85	2.89	3.35	4.97	91.3	2.5	34.5
POLAR	baseline	On	On	Off	3.178	2.72	1.98	3.12	3.14	73.8	58.8	31.8
POLAR	baseline	On	On	On	3.182	2.73	1.97	3.12	3.14	56.3	21.3	31.2
POLAR	weak	Off	Off	Off	3.333	2.82	2.86	3.28	5.35	88.8	0.0	39.0
POLAR	weak	Off	Off	On	3.348	2.90	3.76	3.38	5.24	93.8	0.0	37.9
POLAR	weak	Off	On	Off	3.179	2.74	1.95	3.12	3.14	70.0	51.3	36.2
POLAR	weak	Off	On	On	3.187	2.74	1.99	3.13	3.15	41.3	41.3	36.2
POLAR	weak	On	Off	Off	3.335	2.82	3.47	3.28	5.56	91.3	0.0	32.8
POLAR	weak	On	Off	On	3.353	2.82	2.58	3.32	4.81	96.3	1.3	32.8
POLAR	weak	On	On	Off	3.178	2.74	1.95	3.12	3.14	33.8	32.5	30.7
POLAR	weak	On	On	On	3.183	2.74	1.94	3.13	3.15	81.3	78.8	30.7
POLAR	strong	Off	Off	Off	3.338	2.85	3.79	3.28	5.83	98.8	1.3	37.3
POLAR	strong	Off	Off	On	3.360	2.85	2.99	3.32	5.01	91.3	0.0	37.9
POLAR	strong	Off	On	Off	3.172	2.72	1.96	3.11	3.13	80.0	52.5	35.8
POLAR	strong	Off	On	On	3.181	2.72	1.94	3.12	3.15	76.3	68.8	34.6
POLAR	strong	On	Off	Off	3.337	2.87	3.22	3.28	5.52	83.8	1.3	32.8

Continued on next page

Table 5: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
POLAR	strong	On	Off	On	3.347	2.82	2.64	3.30	4.88	97.5	16.3	32.2
POLAR	strong	On	On	Off	3.179	2.73	1.94	3.12	3.14	86.3	83.8	30.9
POLAR	strong	On	On	On	3.182	2.72	1.94	3.12	3.14	25.0	23.8	30.9
POLAR	discrete	Off	Off	Off	3.338	2.82	2.80	3.29	5.12	93.8	2.5	37.9
POLAR	discrete	Off	Off	On	3.349	2.87	2.90	3.33	5.25	93.8	6.3	37.9
POLAR	discrete	Off	On	Off	3.184	2.75	2.03	3.12	3.14	76.3	73.8	36.4
POLAR	discrete	Off	On	On	3.178	2.71	1.92	3.12	3.15	31.3	18.8	35.8
POLAR	discrete	On	Off	Off	3.333	2.80	4.13	3.28	6.12	97.5	0.0	32.6
POLAR	discrete	On	Off	On	3.351	2.87	2.72	3.33	4.89	85.0	2.5	33.3
POLAR	discrete	On	On	Off	3.189	2.74	1.96	3.13	3.15	87.5	53.8	31.8
POLAR	discrete	On	On	On	3.192	2.74	2.07	3.13	3.15	21.3	26.3	31.7
POLAR	zipfian	Off	Off	Off	3.338	2.85	4.83	3.28	6.55	96.3	11.3	37.9
POLAR	zipfian	Off	Off	On	3.350	2.83	2.58	3.32	4.63	88.8	3.8	39.4
POLAR	zipfian	Off	On	Off	3.171	2.70	1.92	3.11	3.13	52.5	63.8	36.2
POLAR	zipfian	Off	On	On	3.180	2.75	1.95	3.12	3.15	46.3	40.0	35.2
POLAR	zipfian	On	Off	Off	3.340	2.83	3.77	3.28	5.64	98.8	1.3	33.8
POLAR	zipfian	On	Off	On	3.352	2.85	3.13	3.36	5.23	90.0	0.0	33.5
POLAR	zipfian	On	On	Off	3.177	2.72	1.94	3.12	3.14	91.3	67.5	31.8
POLAR	zipfian	On	On	On	3.192	2.74	2.00	3.13	3.15	40.0	26.3	31.8
NOPE	baseline	Off	Off	Off	3.224	2.76	3.71	3.16	6.68	97.5	1.3	41.5
NOPE	baseline	Off	Off	On	3.219	2.75	2.76	3.17	5.49	92.5	10.0	36.5
NOPE	baseline	Off	On	Off	3.140	2.66	2.34	3.09	3.23	97.5	16.3	36.8
NOPE	baseline	Off	On	On	3.150	2.70	2.14	3.09	3.24	81.3	0.0	34.2
NOPE	baseline	On	Off	Off	3.209	2.74	2.99	3.15	6.13	90.0	18.8	32.5
NOPE	baseline	On	Off	On	3.212	2.77	2.90	3.17	5.65	81.3	0.0	30.3
NOPE	baseline	On	On	Off	3.149	2.67	2.35	3.10	3.28	80.0	16.3	30.0
NOPE	baseline	On	On	On	3.147	2.68	2.09	3.09	3.23	88.8	21.3	28.2
NOPE	weak	Off	Off	Off	3.214	2.77	3.10	3.16	6.34	95.0	0.0	40.9
NOPE	weak	Off	Off	On	3.222	2.75	3.05	3.19	5.81	81.3	0.0	36.1
NOPE	weak	Off	On	Off	3.147	2.68	2.07	3.09	3.19	88.8	21.3	37.5
NOPE	weak	Off	On	On	3.142	2.68	2.12	3.08	3.23	82.5	0.0	33.7
NOPE	weak	On	Off	Off	3.215	2.75	4.87	3.16	12.48	95.0	5.0	32.0
NOPE	weak	On	Off	On	3.221	2.73	2.63	3.18	4.84	81.3	0.0	29.3
NOPE	weak	On	On	Off	3.147	2.67	2.48	3.09	3.23	97.5	1.3	30.8
NOPE	weak	On	On	On	3.148	2.68	2.14	3.09	3.24	93.8	1.3	28.4
NOPE	strong	Off	Off	Off	3.231	2.74	4.88	3.18	11.85	90.0	2.5	38.4
NOPE	strong	Off	Off	On	3.214	2.72	2.68	3.17	5.28	82.5	0.0	36.4
NOPE	strong	Off	On	Off	3.144	2.68	2.37	3.09	3.24	96.3	6.3	35.7
NOPE	strong	Off	On	On	3.150	2.70	2.10	3.10	3.25	93.8	7.5	33.8
NOPE	strong	On	Off	Off	3.207	2.72	3.13	3.15	6.20	73.8	0.0	31.4
NOPE	strong	On	Off	On	3.214	2.74	2.49	3.17	4.36	86.3	8.8	28.9
NOPE	strong	On	On	Off	3.146	2.67	2.24	3.09	3.23	83.8	17.5	29.1
NOPE	strong	On	On	On	3.147	2.68	2.16	3.09	3.28	96.3	0.0	27.5
NOPE	discrete	Off	Off	Off	3.202	2.75	3.33	3.15	6.23	85.0	1.3	38.8
NOPE	discrete	Off	Off	On	3.234	2.79	3.11	3.19	5.47	70.0	18.8	37.0
NOPE	discrete	Off	On	Off	3.145	2.68	2.14	3.09	3.27	98.8	16.3	36.0
NOPE	discrete	Off	On	On	3.145	2.68	2.10	3.09	3.25	95.0	3.8	34.3
NOPE	discrete	On	Off	Off	3.212	2.88	3.04	3.16	6.88	42.5	0.0	31.2
NOPE	discrete	On	Off	On	3.211	2.78	2.73	3.16	5.28	82.5	1.3	29.1
NOPE	discrete	On	On	Off	3.145	2.68	2.27	3.08	3.26	91.3	11.3	30.3
NOPE	discrete	On	On	On	3.149	2.68	2.17	3.08	3.32	83.8	0.0	28.6
NOPE	zipfian	Off	Off	Off	3.209	2.71	3.18	3.15	6.40	90.0	0.0	40.8
NOPE	zipfian	Off	Off	On	3.219	2.75	2.71	3.17	5.44	77.5	1.3	36.1
NOPE	zipfian	Off	On	Off	3.141	2.67	2.55	3.08	3.38	96.3	0.0	38.0
NOPE	zipfian	Off	On	On	3.150	2.69	2.11	3.09	3.25	90.0	3.8	33.7
NOPE	zipfian	On	Off	Off	3.207	2.77	3.32	3.15	7.09	82.5	0.0	31.3
NOPE	zipfian	On	Off	On	3.227	2.79	2.91	3.18	5.94	93.8	11.3	30.2
NOPE	zipfian	On	On	Off	3.146	2.68	2.39	3.10	3.25	96.3	21.3	29.6
NOPE	zipfian	On	On	On	3.148	2.68	2.17	3.09	3.28	90.0	6.3	28.7
ROPE	baseline	Off	Off	Off	3.159	2.85	3.36	3.09	5.21	42.5	0.0	40.4
ROPE	baseline	Off	Off	On	3.163	2.86	3.79	3.17	5.86	15.0	0.0	37.2
ROPE	baseline	Off	On	Off	3.145	2.81	2.86	3.09	3.57	73.8	0.0	37.1
ROPE	baseline	Off	On	On	3.151	2.80	2.95	3.12	3.72	13.8	0.0	34.4
ROPE	baseline	On	Off	Off	3.163	3.00	3.47	3.10	5.46	38.8	0.0	33.2
ROPE	baseline	On	Off	On	3.167	2.88	4.06	3.16	5.92	0.0	0.0	30.2
ROPE	baseline	On	On	Off	3.153	2.73	2.45	3.09	3.57	75.0	0.0	31.4
ROPE	baseline	On	On	On	3.154	2.83	2.86	3.12	3.72	32.5	0.0	29.4
ROPE	weak	Off	Off	Off	3.158	2.88	3.46	3.10	5.36	46.3	0.0	41.4
ROPE	weak	Off	Off	On	3.163	2.84	3.96	3.18	5.93	12.5	0.0	37.8
ROPE	weak	Off	On	Off	3.149	2.71	2.46	3.09	3.56	37.5	0.0	36.5
ROPE	weak	Off	On	On	3.156	2.85	2.77	3.12	3.60	23.8	0.0	34.9
ROPE	weak	On	Off	Off	3.155	2.83	3.49	3.10	5.27	38.8	0.0	32.3
ROPE	weak	On	Off	On	3.157	3.05	4.12	3.18	5.86	8.8	0.0	30.5
ROPE	weak	On	On	Off	3.149	2.77	2.76	3.09	3.56	57.5	0.0	31.1
ROPE	weak	On	On	On	3.149	2.77	3.13	3.12	3.80	17.5	0.0	28.0
ROPE	strong	Off	Off	Off	3.165	2.78	3.34	3.10	5.27	30.0	0.0	39.0
ROPE	strong	Off	Off	On	3.169	2.90	3.68	3.19	5.80	31.3	0.0	36.3
ROPE	strong	Off	On	Off	3.158	2.85	2.63	3.10	3.50	51.3	0.0	35.9
ROPE	strong	Off	On	On	3.162	2.86	3.08	3.14	3.76	10.0	0.0	33.2
ROPE	strong	On	Off	Off	3.161	2.85	3.58	3.10	5.28	47.5	0.0	31.1
ROPE	strong	On	Off	On	3.165	2.85	4.03	3.19	5.90	2.5	0.0	29.2
ROPE	strong	On	On	Off	3.155	2.80	2.70	3.10	3.62	61.3	0.0	29.3
ROPE	strong	On	On	On	3.162	2.85	3.08	3.13	3.74	2.5	0.0	28.4
ROPE	discrete	Off	Off	Off	3.165	2.91	3.43	3.11	5.42	22.5	0.0	39.7
ROPE	discrete	Off	Off	On	3.158	2.84	4.05	3.15	5.90	3.8	0.0	36.9
ROPE	discrete	Off	On	Off	3.159	2.83	2.67	3.11	3.56	47.5	0.0	36.3
ROPE	discrete	Off	On	On	3.161	2.79	2.71	3.13	3.74	41.3	0.0	33.7
ROPE	discrete	On	Off	Off	3.171	2.79	3.45	3.12	5.36	51.3	0.0	31.5
ROPE	discrete	On	Off	On	3.164	2.80	3.74	3.16	5.86	21.3	0.0	30.0
ROPE	discrete	On	On	Off	3.162	2.74	2.47	3.11	3.53	50.0	0.0	30.9
ROPE	discrete	On	On	On	3.154	2.79	2.68	3.11	3.67	18.8	0.0	28.9
ROPE	zipfian	Off	Off	Off	3.163	2.82	3.25	3.10	5.22	62.5	0.0	39.8
ROPE	zipfian	Off	Off	On	3.161	2.94	3.94	3.18	5.81	16.3	0.0	37.1
ROPE	zipfian	Off	On	Off	3.149	2.78	2.70	3.09	3.63	46.3	0.0	36.7
ROPE	zipfian	Off	On	On	3.151	2.80	2.63	3.11	3.68	2.5	0.0	33.9
ROPE	zipfian	On	Off	Off	3.161	2.78	3.38	3.10	5.32	47.5	0.0	32.6
ROPE	zipfian	On	Off	On	3.160	3.01	4.11	3.17	5.81	17.5	0.0	30.2
ROPE	zipfian	On	On	Off	3.149	2.73	2.39	3.09	3.55	62.5	0.0	29.8
ROPE	zipfian	On	On	On	3.152	2.79	2.99	3.12	3.78	26.3	0.0	28.5

D Retrieval Breakdown

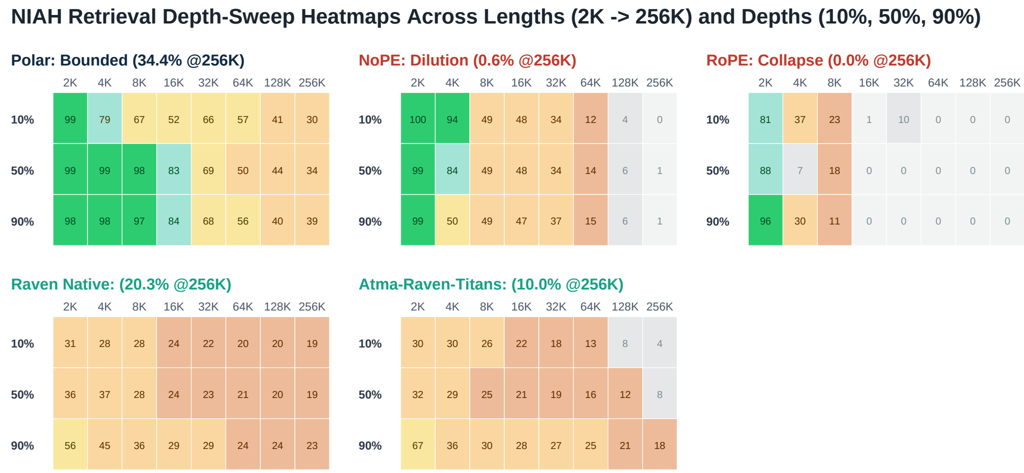

Figure 5: Retrieval depth sweeps for the five promoted models. Each cell is teacher-forced token accuracy.

Model	Haystack	2K	4K	8K	16K	32K	64K	128K	256K
Polar	Synthetic	99.6	98.4	98.5	97.0	94.9	92.3	80.3	68.4
	FinePDFs	98.6	86.3	77.9	49.1	40.7	16.3	3.2	0.3
NoPE	Synthetic	99.5	98.7	98.9	94.9	70.3	26.5	10.2	1.2
	FinePDFs	99.3	52.9	0.0	0.0	0.0	0.0	0.0	0.0
RoPE	Synthetic	81.7	44.9	33.1	0.8	6.8	0.1	0.0	0.0
	FinePDFs	94.8	4.7	0.2	0.0	0.0	0.0	0.0	0.0
Atma-Raven-Titans	Synthetic	64.8	60.7	51.1	46.9	41.7	35.8	25.9	19.3
	FinePDFs	23.2	2.6	0.7	0.2	1.4	0.5	1.6	0.7
Raven Native	Synthetic	55.9	57.7	51.4	48.3	44.3	40.9	38.9	35.1
	FinePDFs	22.1	15.4	8.2	3.3	5.3	2.9	3.7	5.4

Table 6: Zero-shot retrieval accuracy (%) by haystack type and context length, averaged over needle depths.

E Short-Context Quality and Systems

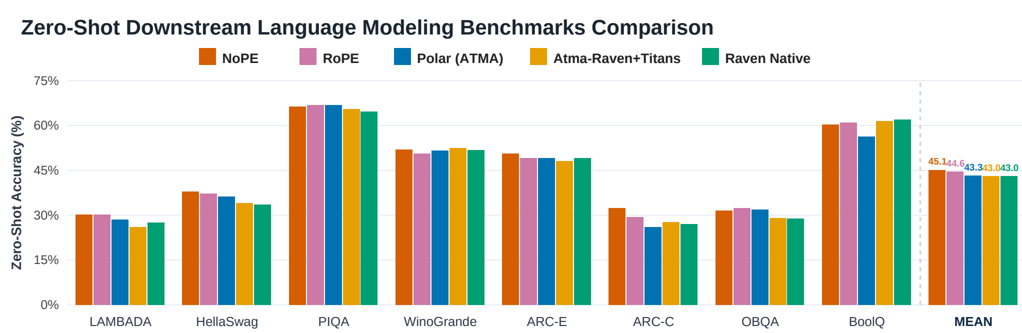

Figure 6: Eight short-context control tasks (22,939 examples per model).

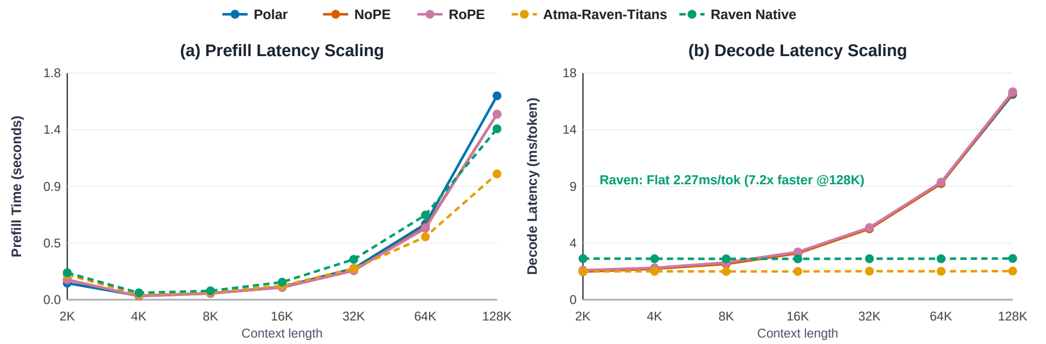

Figure 7: Single-sequence serving on one L40S. Attention decode scales with KV-cache length; recurrent Raven uses fixed state.

Group	Model	Base mean (%)	128K prefill (s)	Decode (ms/token)	Peak alloc. (GiB)	Train MFU (%, 2K)
Matched	NoPE	45.1	1.47	16.4	39.40	42.77
Matched	Polar	43.3	1.62	16.3	39.41	40.44
Matched	RoPE	44.6	1.47	16.5	39.41	42.45
Raven/AdamW	Atma-Raven-Titans	43.1	1.00	2.27	8.62	30.92
Raven/AdamW	Raven Native	43.0	1.36	3.27	6.46	21.53

Table 7: Short-context quality and serving trade-offs. Serving uses one sequence and one timing sample; it is descriptive rather than an uncertainty estimate.

F Stress and Environment Diagnostics

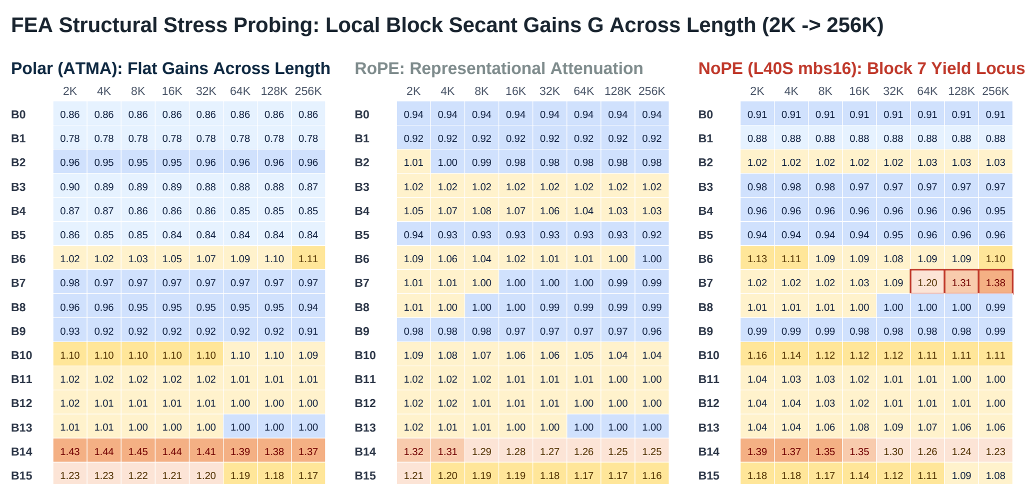

Figure 8: Block-wise sampled secant gains over 64 nested FinePDFs documents and 64 perturbation directions per document. The probe localizes checkpoint-specific changes but does not estimate a global Lipschitz constant.

For each block f_i, the diagnostic samples

$$G_i = \frac{\|f_i(x+\delta x) - f_i(x)\|_{\mathrm{rms}}}{\|\delta x\|_{\mathrm{rms}}}, \qquad \|\delta x\|_{\mathrm{rms}} = 0.02\|x\|_{\mathrm{rms}}. \tag{8}$$

The maximum sampled gain is a lower bound on the local operator norm along sampled directions; it is not a tight bound on the block Lipschitz constant. In the observed NoPE collapse, changes first concentrate near blocks 6–7 and later propagate through the residual stream. Polar's sampled gains and activation moments remain comparatively flat through 256K.

The environment diagnostic re-evaluates checkpoints under PyTorch 2.12 and 2.13, compares microbatch 4 and 16 on L40S, and uses deterministic short-context stream fixtures. These controls rule out evaluation-version and microbatch-only explanations in the tested runs. They do not isolate a unique low-level cause, because the L4/L40S intervention was not replicated over multiple training seeds and kernel configurations.